Lindenbaum's lemma: extending consistent sets to complete ones

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the two tagrefs blocks (provability-exhaustive, proves-compact) name the per-proof-system sources of the two cited facts and are delivered as the author wrote them

Editorial note. The label lem:lindenbaum names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Exhaustiveness of derivability. For any Gamma and A, either Gamma union {A} or Gamma union {not-A} is consistent (the derivability relation is exhaustive).. The contradiction horn in the consistency induction: if both extensions were inconsistent, Gamma_n itself would be, against the induction hypothesis.
- Compactness of derivability. If every finite subset of Gamma is consistent then Gamma is consistent.. Lifts the per-stage consistency to the union Gamma* via its finite subsets.

Lemma 0.1 (Lindenbaum's Lemma)**.** *Every consistent set* Γ *in a language* $\mathcal{L}L$ *can be extended to complete and consistent set* Γ^**.*

Proof. Let Γ be consistent. Let A_0, A_1, … be an enumeration of all the sentences of $\mathcal{L}L$. Define $\Gamma_0 = \Gamma$, and

$$\Gamma_{n+1} = \begin{cases} \Gamma_n \cup \{A_n\} & \text{if } \Gamma_n \cup \{A_n\} \text{ is consistent;} \\ \Gamma_n \cup \{\neg A_n\} & \text{otherwise.} \end{cases}$$

Let $\Gamma^* = \bigcup_{n \geq 0} \Gamma_n$.

Each Γ_n is consistent: Γ_0 is consistent by definition. If $\Gamma_{n+1} = \Gamma_n \cup \{A_n\}$, this is because the latter is consistent. If it isn't, $\Gamma_{n+1} = \Gamma_n \cup \{\neg A_n\}$. We have to verify that $\Gamma_n \cup \{\neg A_n\}$ is consistent. Suppose it's not. Then *both* $\Gamma_n \cup \{A_n\}$ and $\Gamma_n \cup \{\neg A_n\}$ are inconsistent. This means that Γ_n would be inconsistent by prfAX/fol:axd:prv:prop:provability-exhaustive, prfSC/fol:seq:prv:prop:provability-exhaustive, prfND/fol:ntd:prv:prop:provability-exhaustive, prfTab/fol:tab:prv:prop:provability-exhaustive, contrary to the induction hypothesis.

For every n and every $i < n$, $\Gamma_i \subseteq \Gamma_n$. This follows by a simple induction on n. For $n = 0$, there are no $i < 0$, so the claim holds automatically. For the inductive step, suppose it is true for n. We show that if $i < n + 1$ then $\Gamma_i \subseteq \Gamma_{n+1}$. We have $\Gamma_{n+1} = \Gamma_n \cup \{A_n\}$ or $= \Gamma_n \cup \{\neg A_n\}$ by construction. So $\Gamma_n \subseteq \Gamma_{n+1}$. If $i < n + 1$, then $\Gamma_i \subseteq \Gamma_n$ by inductive hypothesis (if $i < n$) or the trivial fact that $\Gamma_n \subseteq \Gamma_n$ (if $i = n$). We get that $\Gamma_i \subseteq \Gamma_{n+1}$ by transitivity of $\subseteq$.

From this it follows that Γ^* is consistent. Here's why: Let $\Gamma' \subseteq \Gamma^*$ be finite. Each $B \in \Gamma'$ is also in Γ_i for some i. Let n be the largest of these. Since $\Gamma_i \subseteq \Gamma_n$ if $i \leq n$, every $B \in \Gamma'$ is also $\in \Gamma_n$, i.e., $\Gamma' \subseteq \Gamma_n$, and Γ_n is consistent. So, every finite subset $\Gamma' \subseteq \Gamma^*$ is consistent. By prfAX/fol:axd:ptn:prop:proves-compact, prfSC/fol:seq:ptn:prop:proves-compact, prfND/fol:ntd:ptn:prop:proves-compact, prfTab/fol:tab:ptn:prop:proves-compact, Γ^* is consistent.

Every sentence of $\mathrm{Frm}(\mathcal{L})$ appears on the list used to define Γ^*. If $A_n \notin \Gamma^*$, then that is because $\Gamma_n \cup \{A_n\}$ was inconsistent. But then $\neg A_n \in \Gamma^*$, so Γ^* is complete. □